भू-दृष्टि (Bhu-Drushti)

उपग्रह-जमीन समन्वय आणि पीक पडताळणी प्रणाली

Satellite-Ground Reconciliation & Crop Field Verification System for Maharashtra

Hackathon	Pune International Agri Hackathon 2026
Theme	Theme 8 - Geospatial & GIS-Based Crop Survey System
Event Dates	15–17 May 2026
Venue	College of Agriculture, Sinchan Nagar Gate, Pune 411005
Product Name	Bhu-Drushti
Team	Aditi Hatwalne - Full-stack developer; GEE satellite pipeline, Python, PostGIS, React Native, Node.js (Express). Yamini Girey - Product and domain research; Maharashtra agriculture policy, GIS analysis, district dashboard design.
Technology Advisors	Anand Hatwalne - 25+ years building software products. Anand Sathe - 30+ years building software products.
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Document Version	3.0 (AEO Administrative Context, App Screens & Functionality)

5.1 समस्या विवरण | Problem Description

Maharashtra conducts the Girdawari, a seasonal crop survey that records the crops sown on every farm plot across 2.3 crore plots statewide. This survey is the legal basis for five critical government functions: subsidy disbursement, PMFBY crop insurance, disaster compensation, MSP procurement planning, and Kisan Credit Card lending. Errors or fraud in this data flow downstream into every one of these decisions.

The entire Girdawari is executed manually today. Satellite data that could verify it already exists and is updated every 5 days, but it is never used for cross-checking or directing field officers to the plots that actually require inspection. Bhu-Drushti builds this missing verification layer.

5.1.1 विशिष्ट समस्या | Specific Pain Points

- AEO overload with no prioritisation: In Maharashtra, an Agriculture Extension Officer (AEO), locally referred to as the Krishi Vistar Adhikari, operates at the Circle level, covering a defined cluster of villages within a Taluka. Each AEO supervises 6 to 10 Krishi Sahayaks (Agriculture Assistants), and since each Krishi Sahayak is responsible for 800 to 1,200 farming families, a single AEO is accountable for the agricultural welfare of approximately 5,000 to 12,000 farming families per season. Without a satellite-assisted prioritisation tool, every plot is treated the same. At 15 minutes per plot, the full verification assignment is physically impossible within the available seasonal window.
- No cross-validation between satellite observation and declared data: Satellite crop classification and farmer declarations are completely separate data streams with no automated connection. An officer has no way to know whether what the satellite detects matches what was declared, which is precisely where fraudulent declarations go undetected.
- No real-time district intelligence: State-level crop maps exist, but District Collectors and Agriculture Officers need taluka and circle-level operational data for subsidy allocation, insurance preparation, and disaster management. This spatial intelligence does not exist as a usable real-time tool anywhere today.
- Crop stress and disaster response are reactive and slow: A drought or flood affecting thousands of plots may not reach a formal report until weeks after the event. There is no automated early warning from satellite data, and no system to produce damage assessments faster than manual AEO surveys.

5.2 संकल्पना विवरण | Concept Statement

> *An Agriculture Extension Officer in Baramati taluka begins the post-Kharif verification window in October. She oversees a Circle comprising six villages. She supervises eight Krishi Sahayaks who together verify 3,200 plots physically. At 15 minutes per plot, this is over 100 working days of combined field work, impossible to complete within the allotted window, even with a full team. She has no tool to direct her team to the plots that actually need a visit, no visibility into which of her Krishi Sahayaks are working in the field today, and no way to report progress to the Taluka Agriculture Officer without assembling data manually.*
>
> *With Bhu-Drushti, she arrives at work and opens a prioritised queue on her phone. It shows 4 plots, the ones the satellite reconciliation engine has flagged as uncertain or suspicious, sorted by severity. She visits the highest-risk plots first. Within the verification window, all flagged plots are confirmed with geo-tagged field evidence, and the remaining auto-confirmed plots carry a satellite audit trail. She is no longer the human bottleneck.*

Bhu-Drushti delivers this through three integrated components:

- A satellite data pipeline that processes Sentinel-2 imagery for Maharashtra every 5 days, computing crop health per plot automatically, using freely available ESA satellite data and Google Earth Engine.
- A reconciliation and risk scoring engine that compares what the satellite detects against farmer declarations, scores each plot by discrepancy, and classifies it as auto-confirmed, flagged for AEO visit, or escalated for senior review.
- An officer-facing system, an offline-first AEO mobile app and a district intelligence dashboard that turns satellite analysis into operational tools for field verification, crop area reporting, and disaster management.

Theme 8 Coverage Verification

Theme 8 requires a system that leverages satellite imagery, drone surveys, GIS mapping, mobile-based geo-tagged field data collection, crop area estimation, crop classification, health monitoring, yield forecasting, and actionable spatial intelligence for government authorities. Bhu-Drushti's coverage against each requirement is as follows.

Theme 8 Requirement	Coverage	Bhu-Drushti Implementation
Satellite imagery	**Phase 1 MVP**	Sentinel-2 via Google Earth Engine. 10m resolution, 5-day revisit. NDVI per plot per pass.
GIS mapping	**Phase 1 MVP**	PostGIS spatial database, Leaflet.js interactive district map, plot polygons from DILRMP/Mahabhunaksha cadastral data.
Mobile geo-tagged field data collection	**Phase 1 MVP**	React Native AEO app with GPS accuracy gate, mandatory geo-tagged photo, offline SQLite storage, and auto-sync queue.
Crop classification	**Phase 1 MVP**	NDVI range matching (Phase 1). Dynamic Time Warping on a full 36-pass time-series curve with >84% accuracy (Phase 2+).
Disaster management & spatial intelligence for government	**Phase 1 MVP**	District dashboard with role-gated access for Collector, Tahsildar, and Agriculture

		Officer. NDVI change detection for disaster damage (simulated demo in Phase 1). Risk summary and escalation queue for actionable decision-making.
Crop area estimation	**Phase 2**	Signal 2: pixel-count area check (NDVI > 0.3 active pixels inside plot boundary converted to hectares). Requires full season time-series. Phase 1 design concept only.
Crop health monitoring	**Phase 2**	NDVI stress zone heatmap on district dashboard. Plots with sudden mid-season NDVI decline flagged automatically. Requires continuous pipeline data.
Drone surveys	**Phase 3**	Not in Phase 1 or Phase 2 scope. Phase 3 extension: drone-captured imagery for plots where satellite cloud cover persistently exceeds 40%, providing sub-metre resolution ground truth for disputed escalated plots.
Yield forecasting	**Phase 3**	Phase 3: NDVI time-series peak and integral values correlated with historical yield records per crop type and agro-climatic zone. Regression model produces plot-level yield estimate at season end.

5.3 गृहितके | Assumptions, Dependencies & Constraints

5.3.1 Critical Dependencies

Dependency	Impact	Mitigation
ePik Pahani Data Access	Stage 2 risk scoring requires data declared by farmers for crop-match validation (Signal 1 and Signal 3). Without ePik Pahani integration, only Signal 2 (area check via satellite) is possible.	Phase 1 uses synthetic declarations. Phase 2 requires a formal MoU with Revenue Dept. via DIPEX/i4C route.
DILRMP Cadastral Shapefiles	Plot boundary polygons required for per-plot NDVI computation. Without shapefiles, satellite pipeline cannot operate at scale.	Phase 1 uses 10 manually digitised plots from Mahabhunaksha. Phase 2 requires data sharing agreement.
Google Earth Engine Quota	Free research tier covers Phase 1 and Phase 2. Phase 3 at full Maharashtra scale may require commercial licence.	ISRO Bhuvan API is alternative for India-specific imagery at no cost. Cost modelling during Phase 2 pilot.

5.3.2 Technical Assumptions

- Human sign-off is mandatory at every decision point. Bhu-Drushti recommends Confirm, Flag, or Escalate. No plot status changes without an authorised officer's action. The system is an intelligence layer, not an autonomous decision-maker.
- The technology stack is indicative and may change based on team expertise and hackathon constraints. The core architecture (satellite pipeline → risk engine → field tools) remains fixed; implementation technologies are flexible.
- Phase 1 prototype uses 10 farm plots in Malegaon Khurd village, Baramati taluka. Plot boundaries are manually digitised from the Mahabhunaksha portal. Farmer declaration data is synthetic, designed to exercise clean, flagged, and escalated outcomes.
- Satellite imagery and compute costs are zero for Phase 1 and Phase 2. Sentinel-2 is free via ESA Copernicus. Google Earth Engine research quota covers the prototype and pilot scale.
- Appendix data, prototype screens, and technology stack are all indicative and subject to change: the satellite NDVI values in Appendix A are real Kharif 2024 data for Malegaon Khurd, but the synthetic farmer declarations in Appendix B are designed solely to demonstrate system logic and do not reflect actual field conditions. The AEO mobile app screens shown in Appendix D and the district dashboard screens shown in Appendix F represent current design intent only; screen layouts, navigation flows, labels, and feature set will evolve during development and may differ from the shipped prototype. The technology stack described in Appendix C and in Phase delivery bullets is indicative: for Phase 1 MVP the backend uses Node.js (Express) instead of the originally planned FastAPI; similar substitutions may occur in other phases based on team expertise, available libraries, and hackathon constraints. The core architecture (satellite pipeline → risk engine → field tools) remains fixed; implementation technologies are flexible. Any specific numbers, counts, or visual designs in these appendices should be treated as illustrative, not as final commitments.

5.3.3 Phase-wise Signal Availability

IMPORTANT: Stage 2 risk scoring depends on external data availability. This table shows what can be promised in each phase.

Signal	Description	Phase 1 (MVP)	Phase 2+
Signal 1 - Crop Match (30%)	Does satellite-detected crop match declared crop?	YES. NDVI range matching with synthetic declarations	DTW on real ePik Pahani data
Signal 2 - Area Check (40%)	Is declared area consistent with satellite-detected active vegetation?	NO. Requires full season time-series data, not available for MVP demo	Pixel-count area check with full season data
Signal 3 - GPS Validation (30%)	Was photo taken inside plot boundary?	DEMO ONLY. Logic demonstrated with AEO app GPS, no real declaration GPS	Real ePik Pahani GPS validation

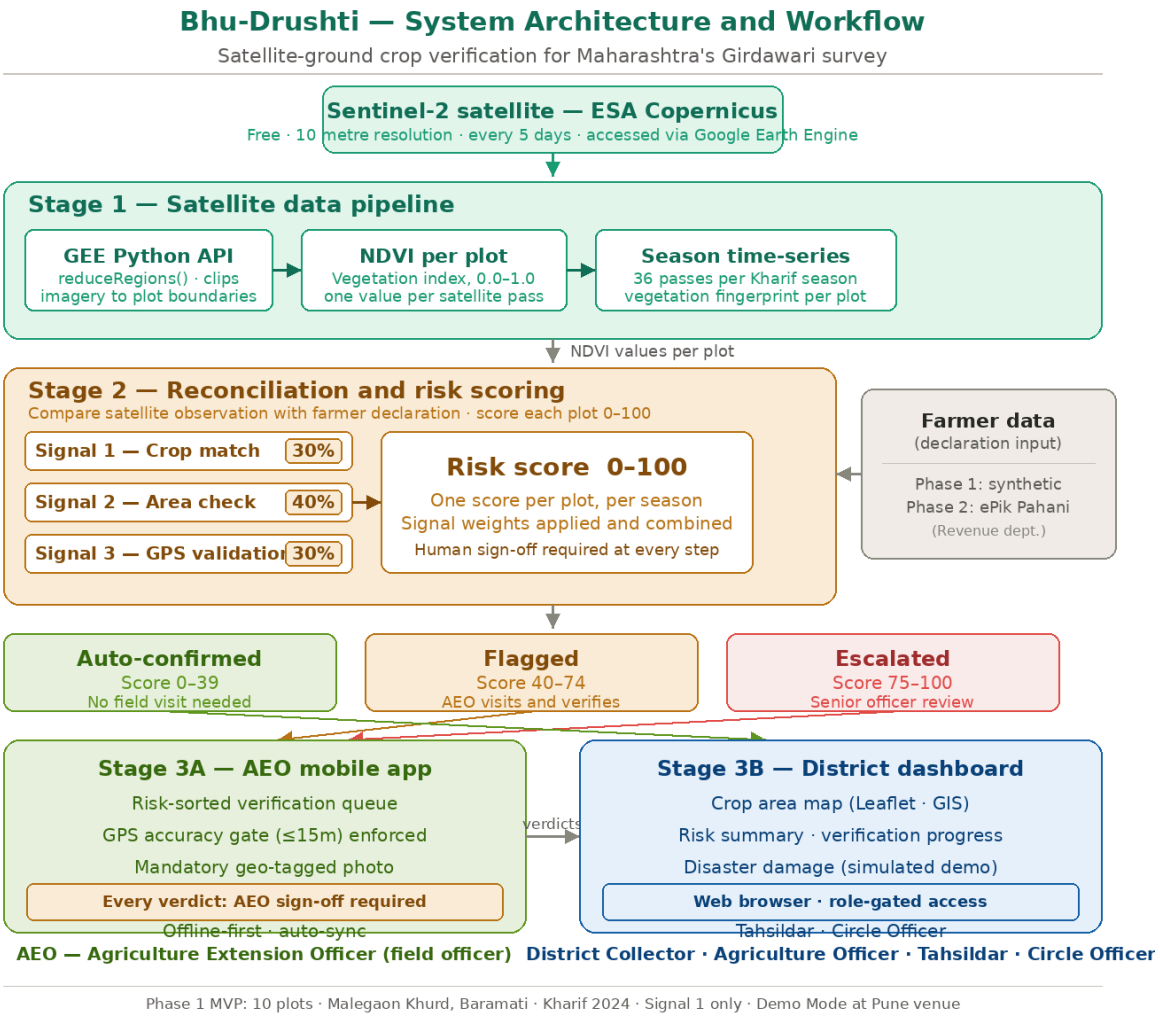
> *MVP Phase 1 Promise: Only Signal 1 (crop match via NDVI range) is fully operational with synthetic data. Signal 2 area check is shown as a design concept. Signal 3 GPS validation is demonstrated through the AEO app capture flow but cannot validate against real ePik Pahani coordinates without the data access.*

5.4 कार्यपद्धती | Methodology

Bhu-Drushti works in three sequential stages. Stage 1 runs continuously during the crop season. Stage 2 runs four times per year. Stage 3 is always live. It is what AEOs and district officers interact with every day.

System Architecture Overview

STAGE 1 Satellite Pipeline	→	STAGE 2 Risk Scoring Engine	→	STAGE 3A AEO Mobile App	+	STAGE 3B District Dashboard
Sentinel-2 imagery → NDVI per plot → Time series		Satellite data + Declarations → Risk score 0-100		Flagged queue + Field capture + Offline first		Crop maps + Statistics + Reports



Stage 1 - उपग्रह डेटा पाइपलाइन | Satellite Data Pipeline

- A European Space Agency satellite (Sentinel-2) passes over Maharashtra every 5 days, imaging every farm plot at 10-metre resolution. This data is free and accessed through Google Earth Engine. No satellite hardware or raw image download is required.

- For each satellite pass, Google Earth Engine clips the satellite image to every plot polygon and computes the mean NDVI (vegetation greenness index) for that plot. One number per plot per pass is stored, not raw pixels.
- Passes where cloud cover exceeds 40% of a plot are automatically excluded. Plots with fewer than 8 valid passes in a season are flagged for mandatory AEO visit regardless of other signals.
- Over the Kharif season (June to November), approximately 36 satellite passes build a time-series curve per plot, a vegetation fingerprint that shows how the crop grew, peaked, and matured.

Stage 2 - समन्वय इंजिन | Reconciliation and Risk Scoring Engine

Stage 2 compares what the satellite observed against what was declared, and produces a 0–100 risk score per plot.

Signal	What it checks	Phase 1 Method	Phase 2+ Method
Signal 1 - Crop match (30%)	Does satellite-detected crop match declared crop?	NDVI range matching: NDVI below 0.25 = fallow regardless of declaration.	Dynamic Time Warping (DTW) on full 36-pass time-series curve. Proven >84% accuracy.
Signal 2 - Area check (40%)	Is declared area consistent with satellite?	NOT AVAILABLE in MVP - requires full season data pipeline.	Count active vegetation pixels (NDVI > 0.3) inside plot boundary. Convert to hectares.
Signal 3 - GPS validation (30%)	Was photo taken inside plot boundary?	Demonstrated with AEO Mode B GPS capture. No real declaration GPS available.	Farmer declaration GPS from data sharing agreement. PostGIS ST_Within check.

Risk score thresholds determine the outcome for each plot:

Risk Score	Status	Meaning	AEO Action
0 – 39	Auto-confirmed	Satellite confirms declaration. No discrepancy.	No visit needed. Logged with satellite evidence.
40 – 74	Flagged	Discrepancy detected - area, crop, or GPS.	AEO visits, confirms or corrects.
75 – 100	Escalated	Significant mismatch - potential fraud.	Circle Officer review. Priority AEO visit.

Stage 3A - AEO मोबाइल अॅप | AEO Mobile App

The AEO app is the operational interface for field officers. It is designed for use outdoors, on budget Android phones, with patchy or no internet connectivity. It runs in different modes depending on time of year; the server controls which mode is active.

Mode	Season	Phase	What the AEO sees and does
Mode B - Verification queue	Post-harvest : Oct–Dec (Kharif), Mar–Apr (Rabi)	Phase 1 MVP	**Risk-sorted verification queue** [Phase 1] - 4 plots sorted by risk score, colour-coded Red (escalate) / Amber (flag). Satellite vs. declared crop shown per row. **Plot polygon on offline map** [Phase 1] - Flagged plot boundary rendered as coloured overlay on pre-loaded OSM map. Live GPS dot shows device position relative to boundary. **GPS accuracy gate** [Phase 1] - Capture blocked until GPS accuracy ≤15m. Ensures photo is taken physically near the plot. **GPS plot auto-ID** [Field only / Demo Mode at venue] - In real field use, GPS auto-identifies the plot the AEO is standing in. At the Pune venue, a clearly labelled Demo Mode allows tapping any plot to simulate a visit - GPS location gate bypassed, all other logic intact. **Mandatory geo-tagged photo** [Phase 1] - Photo required before verdict. GPS coordinates embedded in EXIF. Cannot submit verdict without photo. **Verdict capture (Confirm / Correct / Escalate)** [Phase 1] - Three-action verdict; no other data entry. Saved to device SQLite immediately. **Offline-first / auto-sync** [Phase 1] - 10 plots pre-loaded; full field day with zero internet. Verdicts queued and auto-uploaded on connectivity. **Marathi-first UI** [Phase 1] - All labels, crop names, and error messages in Marathi. Font pre-bundled. **Single-village scope (10 plots)** [Phase 1] - Prototype covers Malegaon Khurd only. Multi-village Circle navigation is Phase 2. **KS team visibility** [Phase 2] - Requires multi-user infrastructure not in Phase 1 prototype. **TAO / BDO report generation** [Phase 2] - Requires multi-village full-season data. Not deliverable in Phase 1.
Mode A - Sowing capture	Sowing season: Jun–Jul (Kharif), Oct–Nov (Rabi)	Phase 2	GPS auto-detects which plot the AEO is standing in. Survey number and owner name fill automatically from the local database - no manual typing. AEO selects crop from Marathi dropdown, enters area, takes photo. Produces Bhu-Drushti's own verified ground truth, reducing dependence on third-party declaration data.

App Design Principles - What the App Promises and Delivers

- Offline-first architecture: all required data pre-loaded onto the device each morning over WiFi. The entire field day - map, plot boundaries, risk queue, photo capture, verdict recording- works with zero internet. Sync happens automatically in the evening when any connectivity is detected.
- Plot polygon on screen: each flagged plot is rendered as a coloured polygon overlay on an offline map using react-native-maps. As the AEO approaches, a live GPS dot shows their position relative to the boundary.

- No manual data entry: the AEO never types a survey number. GPS auto-identifies the plot. Crop is selected from a dropdown. The area is validated against the official plot area from the cadastral records.
- Data integrity: every field action is saved to the device SQLite database immediately. A pending upload queue retries automatically on connectivity. No data is lost even if the app crashes.
- Language: Marathi is the primary language on every screen - crop names, button labels, error messages, status indicators.
- Circle-first navigation hierarchy (Phase 2): the AEO's primary context is their Circle - a defined cluster of villages within a Taluka. Navigation flows from Circle → Village → Plot, matching the AEO's natural mental model. Phase 1 covers a single village (Malegaon Khurd) and does not exercise multi-village Circle navigation. This hierarchy becomes operational in Phase 2 when full taluka plot data is loaded.
- KS team visibility (Phase 2): the AEO supervises 6 to 10 Krishi Sahayaks. The app will provide a dedicated team view showing each KS officer's plot count, progress percentage, and last-sync timestamp, with automatic flagging of non-reporting staff. This feature requires KS officer accounts, server-side activity tracking per officer, and multi-user data infrastructure that is not part of the Phase 1 single-AEO prototype.
- Dual reporting support (Phase 2): the AEO reports to both the Taluka Agriculture Officer (TAO) for technical agricultural data and the Block Development Officer (BDO) for administrative matters. The Reports screen is designed to generate both outputs in one tap from verified field data. This requires multi-village, full-season data to produce a meaningful report and is a Phase 2 deliverable once taluka-level data is available.

Stage 3B - जिल्हा डॅशबोर्ड | District Intelligence Dashboard

The district dashboard is a web application for District Collectors, Agriculture Officers, Tahsildars, and Circle Officers. It is role-based. Each officer sees only the scope and actions appropriate to their position.

MVP Dashboard Scope (Phase 1)

The MVP dashboard is deliberately limited to demonstrable functionality with 10 prototype plots. Full-scale features require Phase 2 data access.

Dashboard Panel	Audience	What it shows	Phase
Crop area map	All officers	Interactive Leaflet map with plot polygons coloured by satellite-classified crop type. Click any plot for full risk breakdown.	Phase 1 MVP
Risk summary	Tahsildar, Agri Officer	Counts of auto-confirmed, flagged, escalated, and pending plots. Simple progress tracker.	Phase 1 MVP
Disaster damage concept	District Collector	SIMULATED DEMO: NDVI change detection for a declared event area. Shows concept with synthetic pre/post data.	Phase 1 MVP (Demo)
Crop area statistics	Collector, Agri Officer	Hectares of each crop by taluka and circle. Season-over-season comparison.	Phase 2
NDVI stress zone alert	Agriculture Officer	Heatmap of plots with sudden NDVI decline mid-season. Requires continuous pipeline data.	Phase 2
Escalation queue	Circle Officer, Tahsildar	All escalated plots with risk scores and AEO notes. Route to senior review.	Phase 2
DES report export	Agriculture Officer	One-click export of district crop statistics in DES Maharashtra format.	Phase 2

5.5 अंमलबजावणीचे टप्पे | Implementation Phases

Phase 1 - MVP Hackathon Prototype (Now to May 2026)

> *End goal: A working prototype for Malegaon Khurd village, Baramati taluka - demonstrating the complete satellite pipeline, Signal 1 risk scoring, AEO app (Mode B), and basic district dashboard with real Sentinel-2 data.*

What Phase 1 WILL Deliver

- Satellite pipeline operational: Sentinel-2 NDVI computed for 10 Malegaon Khurd plots, Kharif 2024. Validated - all 10 plots returned NDVI values between 0.64 and 0.79, consistent with healthy peak Kharif vegetation.
- Plot boundaries: 10 plots in Malegaon Khurd manually digitised from Mahabhunaksha. Survey numbers and owner names from actual public revenue records. Uploaded to Google Earth Engine as a validated FeatureCollection asset.
- Risk scoring (Signal 1 only): NDVI range matching producing 0–100 risk scores. Outputs: 6 auto-confirmed, 3 flagged, 1 escalated from synthetic declarations.
- **AEO mobile app - Mode B** (Phase 1 confirmed features):
  - **Risk-sorted verification queue:** 4 plots sorted by risk score - 1 escalated (Gat 113), 3 flagged (Gat 108, 111, 117). Colour-coded Red/Amber. Each row shows satellite-detected crop, declared crop, and risk score.
  - **Plot polygon on offline map:** Each plot boundary rendered as a coloured overlay on pre-loaded OSM map tiles. A live GPS dot shows device's position relative to the boundary.
  - **GPS accuracy gate:** Verdict capture is blocked until the device's GPS accuracy reaches ≤15 metres.
  - **GPS plot auto-ID - field use:** GPS auto-identifies the plot the AEO is standing in using turf.js point-in-polygon on the pre-loaded cadastral dataset. No manual plot selection or typing needed.
  - *Demo Mode (hackathon venue only): The device is not physically in Baramati during the Pune event, so a clearly labelled Demo Mode lets the judge tap any plot to simulate a field visit. The GPS location gate is bypassed; all other logic - accuracy check, photo capture, verdict, SQLite save, sync queue, runs exactly as in real field use.*
  - **Mandatory geo-tagged photo:** Camera capture required before verdict. GPS coordinates embedded in photo EXIF. Compressed to ~80 KB. The verdict cannot be submitted without a photo.
  - **Verdict capture:** Three buttons - Confirm, Correct, Escalate. Saved to the device SQLite immediately with a timestamp. No other data entry required.
  - **Offline-first / auto-sync:** All 10 plots pre-loaded at morning WiFi sync. Full field day works with zero internet. Pending verdicts are queued and automatically uploaded when connectivity is detected.
  - **Marathi-first UI:** All labels, crop names, error messages, and status indicators in Marathi. The Noto Sans Devanagari font is pre-bundled in the app.
- District dashboard (basic): Leaflet map with plot polygons coloured by satellite crop classification, risk summary panel, simulated disaster damage demo.
- Backend: Node.js (Express) REST API + PostGIS database running locally on Docker. Three API endpoints: plot queue delivery, verdict upload, and NDVI delta for heatmap.

What Phase 1 Will NOT Deliver

Excluded Item	Reason
Signal 2 (Area check)	Requires a full season time-series pipeline with per-pass NDVI data, not just a peak-season snapshot. Will be available in Phase 2.
Signal 3 (Real GPS validation)	Requires ePik Pahani data access with farmer photo GPS coordinates. Not available without an MoU.
Mode A (Sowing capture)	Sowing season is June-July 2026, after the hackathon. Cannot demo live capture during the May event.
Live stress heatmap (Mode C)	Requires a continuous pipeline running throughout the season. MVP uses historical snapshot data only.
DES report export	Requires full district data and formal format compliance. Phase 2 deliverable.
Full taluka scale	Requires DILRMP data access. MVP is limited to 10 manually digitised plots.
Real farmer declaration data	Phase 1 uses synthetic declarations only. Real data requires ePik Pahani MoU.
Circle-first navigation (multi-village)	Phase 1 covers a single village. Circle hierarchy requires multi-village data loading, available in Phase 2.
KS team visibility	Requires KS officer accounts, server-side activity tracking per officer, and multi-user data infrastructure. Not available in the Phase 1 single-AEO prototype.
TAO and BDO report generation	Meaningful TAO and BDO reports require multi-village, full-season verified data. Phase 1 has 10 plots in one village. Report generation is a Phase 2 deliverable.
Live GPS plot auto-identification at the hackathon venue	GPS auto-identification of the plot requires the device to be physically inside a Malegaon Khurd plot in Baramati. The hackathon venue is Pune College of Agriculture - the GPS will report the Pune campus location, not any prototype plot boundary. A clearly labelled Demo Mode is provided: the judge taps any plot to simulate a field visit; the GPS location gate is bypassed while all other capture logic (accuracy check, photo, verdict, SQLite save, sync queue) runs exactly as in real field use. This limitation is inherent to any location-based field app demonstrated off-site and is not a deficiency of the system.

Phase 2 - Taluka Pilot (3–9 months post-hackathon)

> *End goal: Bhu-Drushti deployed and actively used in one full taluka in Pune district, processing both Kharif and Rabi seasons with real plot boundary data and AEO field capture.*

- DILRMP data access: Formal data sharing agreement with Maharashtra Revenue Department for Pune district cadastral data - all agricultural plot boundaries and survey numbers.
- Full time-series pipeline: Mode A GEE pipeline (per-pass full season, 36 passes) replacing Phase 1 peak-season snapshot. Signal 2 upgraded to pixel-count area check.
- Mode A sowing capture live: AEOs use the app during Kharif and Rabi sowing windows to capture crop declarations directly.
- Signal 3 with real GPS: Integration of AEO Mode A GPS coordinates into the Signal 3 calculation.
- DES report export live: District crop statistics in DES Maharashtra format, auto-generated at the end of the season.

Phase 3 - Full Maharashtra Rollout (9–24 months)

> *End goal: Bhu-Drushti operational across all 36 Maharashtra districts and 358 talukas, providing continuous satellite-verified crop intelligence throughout both seasons.*

- All 2.3 crore agricultural plots: Full DILRMP cadastral data via state-level MoU. One-time GEE upload.
- 358-taluka parallel processing: 358 Google Earth Engine tasks submitted simultaneously per satellite pass.
- DTW classification with calibrated reference curves: Crop reference NDVI signatures recalibrated each season from AEO ground truth verifications.
- AgriStack integration: Crop Sown Registry API via UFSI once production access is granted under the Digital Agriculture Mission.

5.7 व्यवहार्यता | Feasibility

5.7.1 तांत्रिक व्यवहार्यता | Technical Feasibility

Phase 1

- Satellite pipeline: Working and validated. GEE Python API script processes Sentinel-2 imagery using reduceRegions() - the same function that processes 10 plots in the prototype processes 2.3 crore plots at full Maharashtra scale without code change.
- Risk scoring: DTW classification is a peer-reviewed technique with published accuracy above 84% for Indian Kharif crops at 10m Sentinel-2 resolution. Phase 1 uses NDVI range matching as a simplified version.
- AEO app: React Native with Expo-SQLite offline storage. On-device polygon geometry via turf.js. GPS accuracy monitoring via expo-location. All libraries are open source.
- No government API dependency in Phase 1: all data is free satellite imagery or manually sourced public cadastral data. The prototype is fully self-contained.

Phase 2 and Phase 3

- DILRMP cadastral data: One-time shapefile upload to GEE after formal MoU. No pipeline code change.
- GEE commercial licence: Required at full Maharashtra scale if the free tier is exceeded. ISRO Bhuvan API is an alternative.
- Infrastructure scale: Cloud SQL + Cloud Run architecture scales linearly. Each additional taluka requires only loading its plot boundaries.

5.7.2 आर्थिक व्यवहार्यता | Economic Feasibility

Phase	Infrastructure	Monthly Cost	Key Cost Driver
Phase 1 - MVP	Local laptop, Docker PostGIS, GEE free tier	₹0	No infrastructure cost
Phase 2 - Pilot (1 taluka)	GCP Cloud SQL e2-small + Cloud Run	~₹1,800	Database VM + API hosting
Phase 3 - Full Maharashtra	Cloud SQL e2-standard-4 + Cloud Run autoscaling	~₹25,000 (~$300)	Database with read replicas

5.8 विस्तारक्षमता | Scalability within Maharashtra

Phase 1 - Single village proof of concept

- 10 plots in Malegaon Khurd village, Baramati. The complete workflow is demonstrated. The pipeline architecture is identical at any scale - only the GEE asset and database size change.
- Configurable crop reference library: adding a new crop requires one new NDVI curve in a JSON file. No code change.

Phase 2 - Taluka and district scale

- Same pipeline, more plots: each additional taluka requires loading its boundaries. No new development.
- Multi-tier access: Talathi, Circle Officer, Tahsildar, and Collector each see a role-appropriate view of the same data.

Phase 3 - Full Maharashtra, then beyond

- 358 talukas × 36 passes = 12,900 GEE tasks per Kharif season, all submitted in parallel. Within GEE research quota.
- Replicable to other states: the core pipeline is state-agnostic. Adapting to another state requires new crop reference curves and cadastral data - estimated 6–8 weeks per state.

5.9 प्रशिक्षण आणि स्वीकृती | Training and Adoption

Bhu-Drushti is designed to require minimal training because it mirrors what field officers already do: observe crops, record findings, review maps.

Training Component	Duration	Approach
AEO mobile app - Mode B	3 hours	Group session at the taluka office. Three actions only: Confirm, Correct, Escalate. No GIS knowledge required.
Offline operation	Built in	App pre-loads all data each night on WiFi. No connectivity needed during field hours.
District dashboard	2 hours	Remote orientation via video call. Map navigation, filter by crop and taluka, and interpret colour coding.
Marathi language throughout	Zero	All screens, labels, crop names, and error messages in Marathi as the primary language.

5.10 संदर्भ | References

- MahaAgri-AI Policy 2025–2029 - Maharashtra Agriculture Department. AI-powered remote sensing and GIS crop survey tools are listed as a funded priority. Initial allocation: ₹500 crore. https://krishi.maharashtra.gov.in
- Sentinel-2 - ESA Copernicus Programme. Free 10m multispectral imagery, 5-day revisit. Accessed via Google Earth Engine. https://sentinel.esa.int
- Google Earth Engine - a planetary-scale geospatial analysis platform. Free for research and non-commercial use. https://earthengine.google.com
- Digital Agriculture Mission - Government of India. ₹2,817 crore sanctioned nationally. Maharashtra is a participating state. https://agricoop.gov.in
- DILRMP - Digital India Land Records Modernisation Programme. National digitised cadastral shapefile repository. https://dilrmp.gov.in
- Mahabhunaksha - Maharashtra Revenue Department cadastral map portal. https://mahabhunakasha.mahabhumi.gov.in
- Pune International Agri Hackathon 2026 - Theme 8: Geospatial & GIS-Based Crop Survey System. https://www.puneagrihackathon.com

5.11 परिशिष्ट | Appendix

Appendix A - Satellite Pipeline Validation (Real Data)

GEE pipeline run for 10 Malegaon Khurd plots, Kharif 2024 peak season. Mode B export - qualityMosaic peak NDVI across all 17 valid satellite passes found for the area.

Survey No	Gat	Area (ha)	Peak NDVI	Interpretation
MH-PUN-MLGK-108	108	0.43	0.761	Active crop - healthy
MH-PUN-MLGK-109	109	0.19	0.785	Active crop - healthy
MH-PUN-MLGK-110	110	0.24	0.639	Active crop - healthy
MH-PUN-MLGK-111	111	0.19	0.784	Active crop - healthy
MH-PUN-MLGK-112	112	0.12	0.641	Active crop - healthy
MH-PUN-MLGK-113	113	0.19	0.688	Active crop - healthy
MH-PUN-MLGK-114	114	0.23	0.759	Active crop - healthy
MH-PUN-MLGK-115	115	0.42	0.767	Active crop - healthy
MH-PUN-MLGK-116	116	0.23	0.697	Active crop - healthy
MH-PUN-MLGK-117	117	0.34	0.658	Active crop - healthy

> *GEE project: bhudrushti. Asset: projects/bhudrushti/assets/dilrmp_plots. All 10 plots returned valid NDVI in the range 0.64–0.79, consistent with healthy agricultural vegetation at peak Kharif season in Baramati.*

Appendix B - Stage 2 Risk Scoring - Prototype Design

10 plots with synthetic farmer declaration data designed to exercise all four risk outcomes:

Survey No	Declared crop	Satellite sees	Error type	Risk output
MH-PUN-MLGK-108	Soybean	Soybean (active)	Area inflated 15%	FLAG
MH-PUN-MLGK-109	Soybean	Soybean (active)	None - clean	AUTO-CONFIRM
MH-PUN-MLGK-110	Cotton	Cotton (active)	None - clean	AUTO-CONFIRM
MH-PUN-MLGK-111	Soybean	Soybean (active)	Area inflated 25% + GPS outside	FLAG
MH-PUN-MLGK-112	Sugarcane	Sugarcane (active)	None - clean	AUTO-CONFIRM
MH-PUN-MLGK-113	Wheat	Fallow (inactive)	Wrong crop + area inflated 30%	ESCALATE
MH-PUN-MLGK-114	Onion	Onion (active)	None - clean	AUTO-CONFIRM
MH-PUN-MLGK-115	Soybean	Soybean (active)	None - clean	AUTO-CONFIRM
MH-PUN-MLGK-116	Sugarcane	Sugarcane (active)	None - clean	AUTO-CONFIRM
MH-PUN-MLGK-117	Cotton	Sugarcane (active)	Wrong crop declared	FLAG

Result: 6 auto-confirmed (60%), 3 flagged (30%), 1 escalated (10%). This distribution validates that the risk engine correctly distinguishes clean declarations from suspicious ones.

Appendix C - Technology Stack (Indicative)

NOTE: *Technology choices are indicative and may change based on team expertise and hackathon constraints. The core architecture remains fixed; implementation technologies are flexible.*

Stage 1 - Satellite Pipeline

Component	Technology	Purpose
Satellite imagery	Sentinel-2 via GEE	Free 10m resolution imagery, 5-day revisit
Compute platform	Google Earth Engine	Planetary-scale raster processing, no server needed
NDVI extraction	GEE Python API	reduceRegions() for batch per-plot NDVI
Transit storage	Google Cloud Storage	CSV staging before database ingest

Stage 2 - Risk Scoring Engine

Component	Technology	Purpose
Database	PostgreSQL + PostGIS	Spatial queries, plot geometry storage
DTW classification	tslearn (Python)	Dynamic Time Warping for crop curve matching
API server	Node.js (Express)	REST endpoints for app and dashboard

Stage 3A - AEO Mobile App

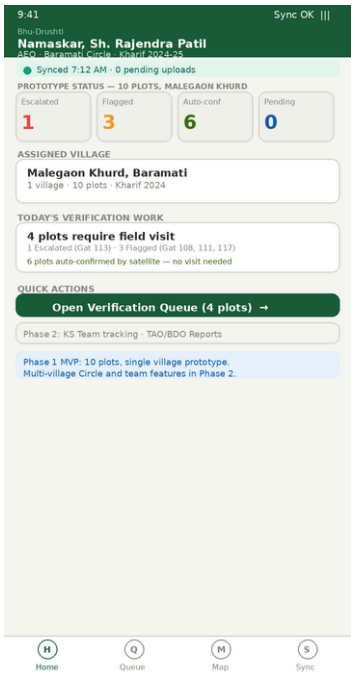
Component	Technology	Purpose
App framework	React Native (Expo)	iOS and Android from one codebase
Map + polygon display	react-native-maps	Plot polygon overlays, AEO GPS dot, offline tiles
On-device geometry	turf.js	GPS point-in-polygon check - runs offline
GPS tracking	expo-location	Continuous accuracy monitoring, capture blocked until ≤15m
Camera	expo-camera + expo-image-manipulator	Photo capture, GPS embedding, compression to ~80 KB
Local database	expo-sqlite	All pre-loaded data and pending upload queue
Language	react-i18next + Noto Sans Devanagari	Marathi-first UI, font pre-bundled

Stage 3B - District Dashboard

Component	Technology	Purpose
Dashboard framework	React + Leaflet.js	Web-based - works in any browser
Charts	Recharts	Crop area bar charts and season comparison
Map tiles	MVT vector tiles via pg_tileserv	Renders any number of plots efficiently
Report generation	WeasyPrint	Server-side PDF in DES format

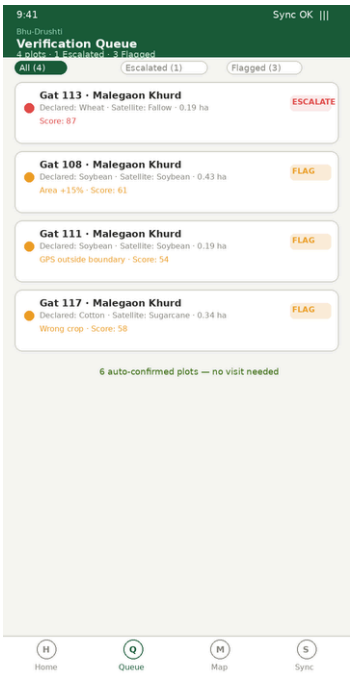
Appendix D - AEO Mobile App Screens (Phase 1 MVP)

These screens represent Phase 1 MVP design intent. Layouts, labels, and content will evolve during development. All screens function fully offline. Marathi is the primary language in the live app; English labels are used here for clarity. KS Team management screen is excluded from Phase 1 MVP as it requires team data infrastructure not available in the prototype.



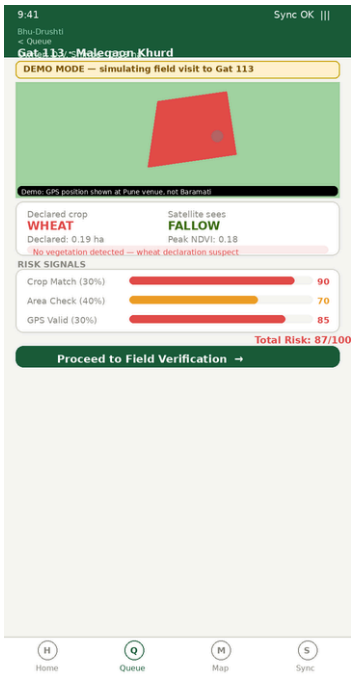
Screen 1 - Home Dashboard

Morning briefing: risk counts, Circle summary, KS team progress, quick-action buttons.



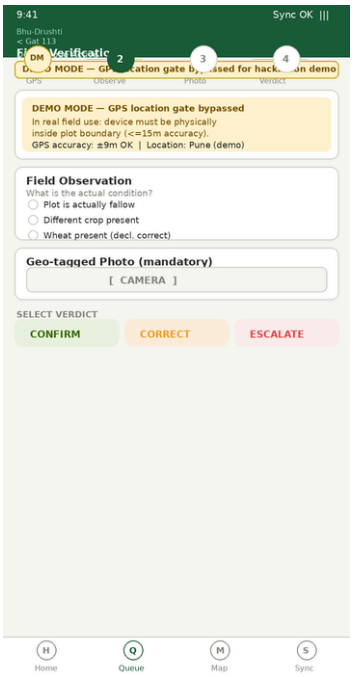
Screen 2 - Verification Queue

Risk-sorted list of all flagged and escalated plots; filter by village or status.



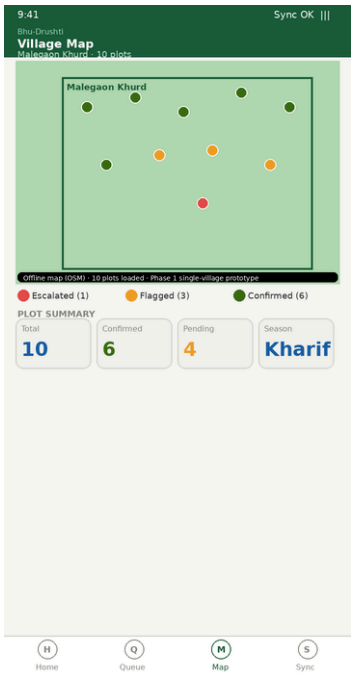
Screen 3 - Plot Detail

Full satellite evidence per plot: declared vs. detected crop, signal scores, and offline map.



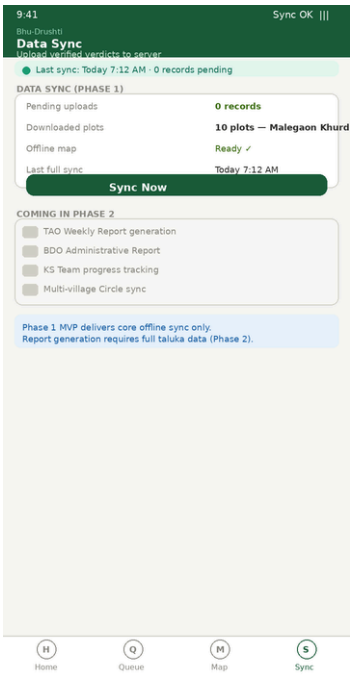
Screen 4 - Field Verification

4-step GPS-gated capture: GPS lock, observation, mandatory geo-tagged photo, verdict.



Screen 5 - Circle Map

Offline village map coloured by risk concentration; tap a village to filter the queue.

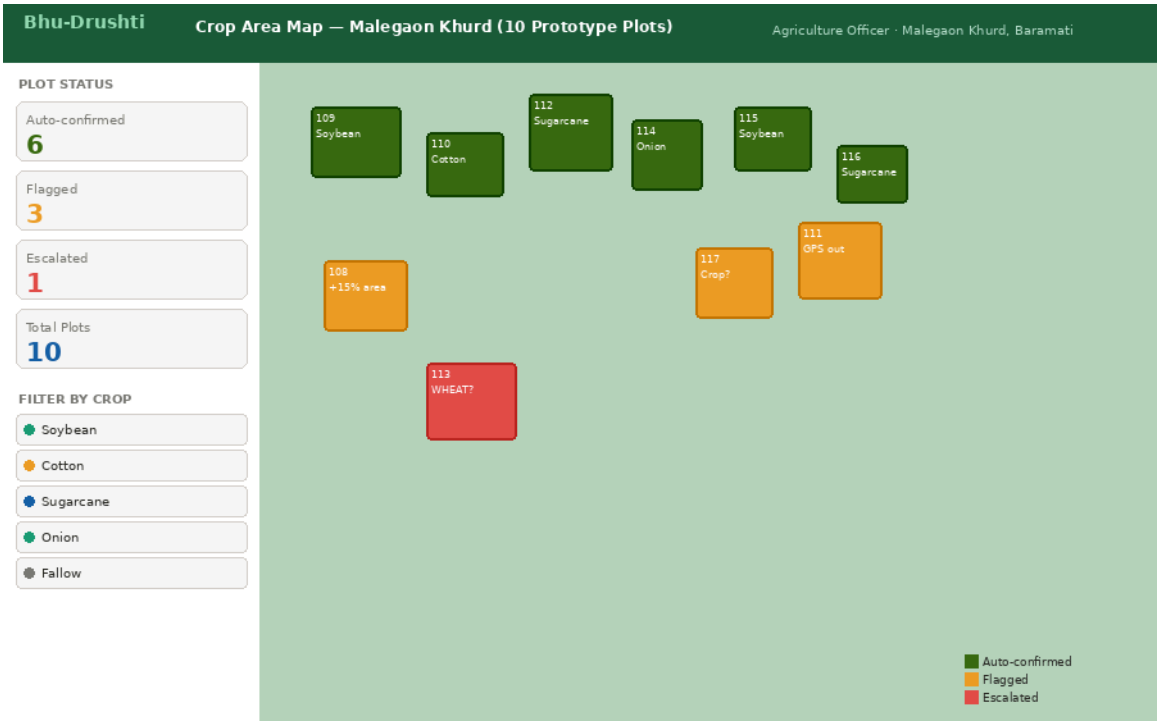


Screen 6 - Reports and Sync

One-tap TAO weekly report and BDO administrative report; data sync status and controls.

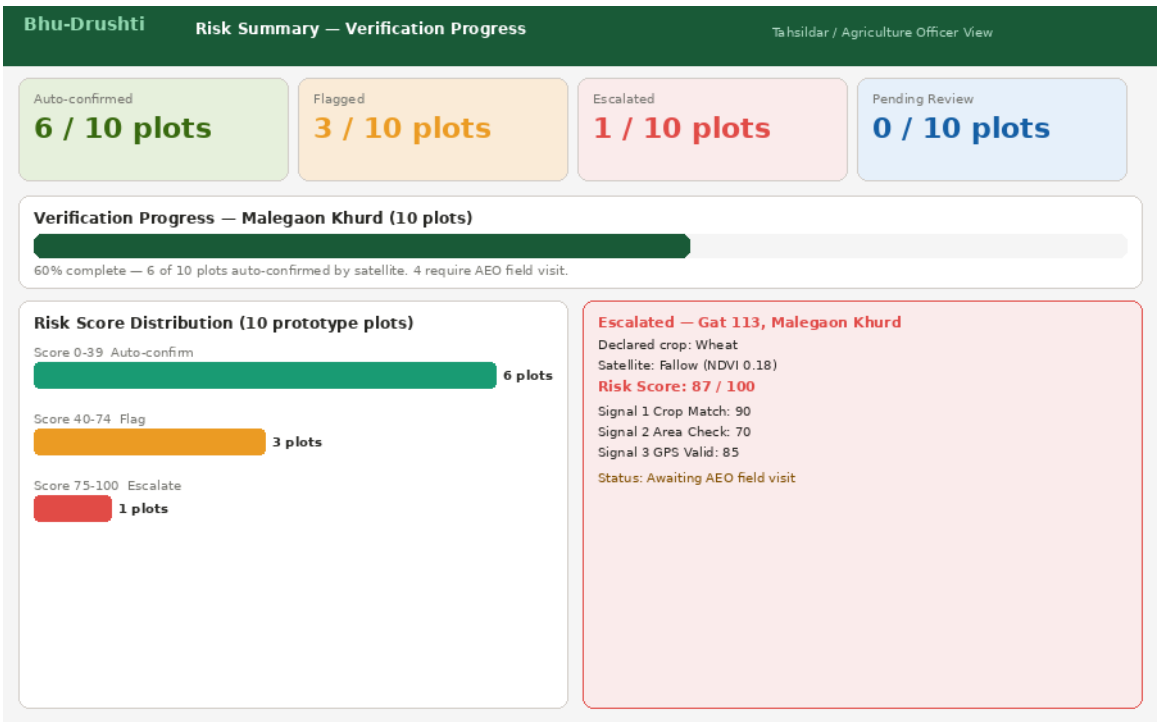
Appendix F - District Intelligence Dashboard Screens (Phase 1 MVP)

These screens show the Phase 1 MVP dashboard scoped to 10 prototype plots in Malegaon Khurd, Baramati. The dashboard is a web application accessible to District Collectors, Agriculture Officers, and Tahsildars. Screen layouts and data will differ in Phase 2 when full taluka data is available.



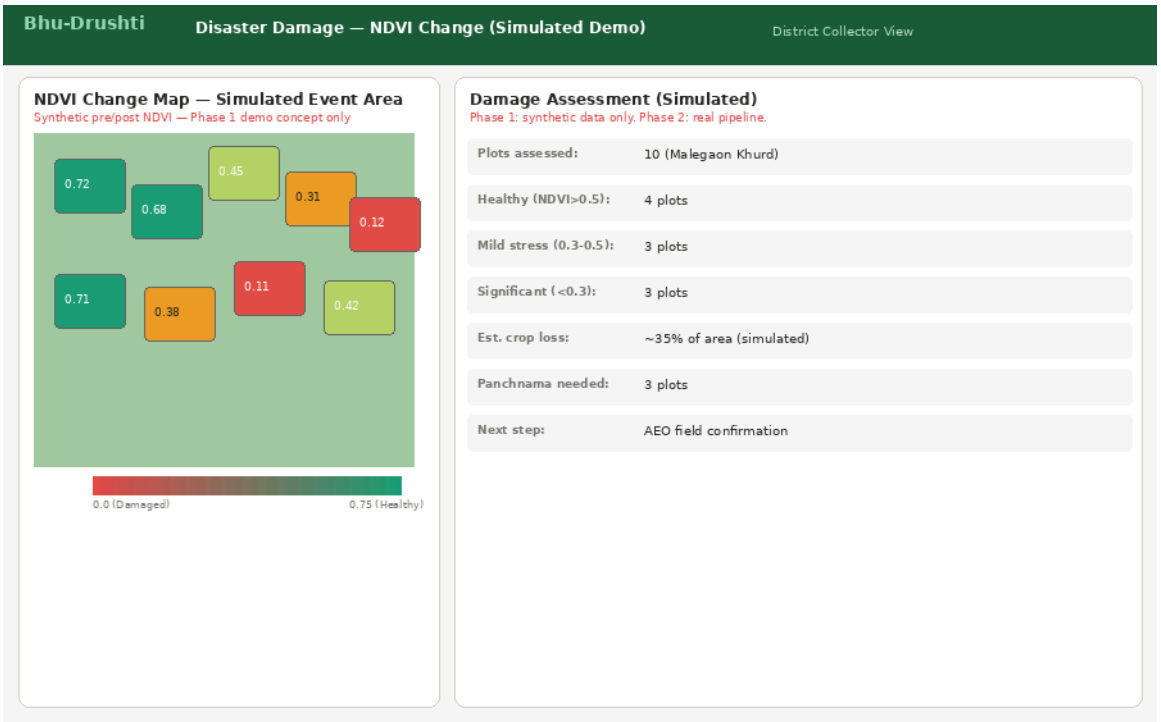
Dashboard Screen 1 - Crop Area Map

Interactive Leaflet map: plot polygons coloured by satellite-classified crop type. Click any plot for full risk breakdown. Available Phase 1 MVP.



Dashboard Screen 2 - Risk Summary Panel

Counts of auto-confirmed, flagged, and escalated plots with an overall verification progress bar and per-plot risk breakdown. Available Phase 1 MVP.



Dashboard Screen 3 - Disaster Damage Module

NDVI change detection showing pre/post event vegetation health per plot. Phase 1 is a simulated demo with synthetic data; real damage assessment requires Phase 2 continuous pipeline.